



St. Catherine's School Waverley
August 2012

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Extension I Mathematics

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11-14
- Task weighting – 40%

Student Number

Total Marks – 70

Section I

Pages 3-6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section
- Answer on the multiple choice answer sheet provided.

Section II

Pages 7-13

60 marks

- Attempt Questions 11-14
- Allow about 1 hour and 45 minutes for this section
- Answer each question in the booklet provided.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Section I

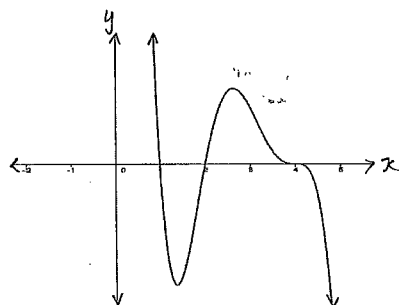
Total marks - 10

Attempt Questions 1-10

All questions are of equal value.

Answer either A, B, C or D on the multiple choice answer sheet provided.

1)



A possible equation for the graph above is:

a) $y = (x - 1)(x - 2)(x - 4)^2$

b) $y = (x - 1)(2 - x)(x - 4)^3$

c) $y = (x - 1)(x - 2)(x - 4)$

d) $y = (x - 1)^2(x - 2)^2(4 - x)$

2) The Cartesian equation of the curve with parameters p, q where

$$x = p + q$$

$$y = p^2 + q^2 + 4pq \text{ and } pq = -1 \text{ is given by:}$$

a) $y = x^2 - 4$

b) $y = x^2 + 2$

c) $y = (x - 1)^2$

d) $y = x^2 - 2$

3) A particle moves in a straight line and its position at time t (in seconds) is given by

$$x = 3 \sin\left(4t + \frac{\pi}{4}\right) + 1 \quad \text{where } x \text{ is measured in metres.}$$

Its maximum speed will be:

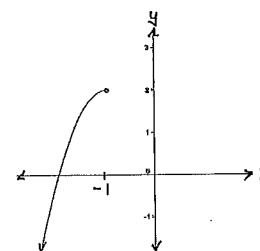
a) 4 m/s

b) 12 m/s

c) 13 m/s

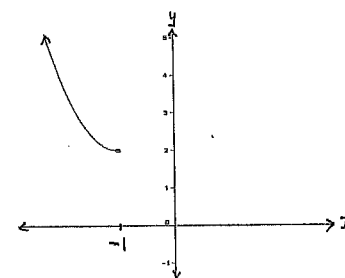
d) unable to be determined.

4)

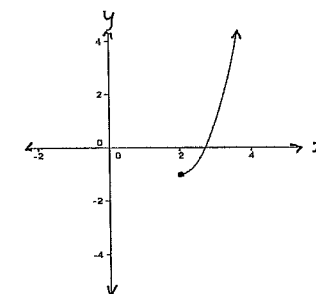


A possible inverse function for the graph shown above is:

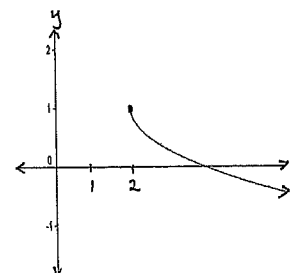
a)



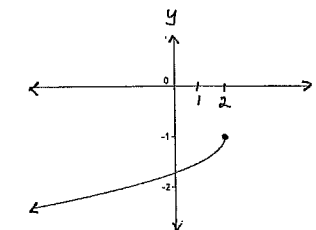
b)



c)



d)



- [illegible]

- a) $\frac{dv}{dt}$
- b) $\frac{dv}{dx}$
- c) $v \frac{dv}{dx}$
- d) $\frac{d(\frac{1}{2}v^2)}{dx}$

-
- A triangle with side lengths labeled as $x+2$, $7-2x$, and $x+1$.

a) $1 < x < 3$

b) $-2 < x < 3\frac{1}{2}$

c) $0 < x < 4$

d) $-1 < x < 3\frac{1}{2}$

Page 6

Section II

Total marks - 60

Attempt Questions 11-14

All questions are of equal value

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Question 11 (15 marks) Start a new booklet

Marks

- a) Find the gradient of the tangent to $y = 3 \cos^{-1} \frac{x}{2}$ at the point where $x = 1$. 2
- b) Use the substitution $u = x^2$ to find $\int \frac{x}{\sqrt{1-x^4}} dx$ 2
- c) A class consists of 9 girls and 6 boys. How many ways are there of selecting a committee of 3 girls and 2 boys from this class? 1
- d) Calculate $\int_0^3 \frac{2}{\sqrt{9-4x^2}} dx$ 3
- e) The interval AB has endpoints $A(-4,6)$ and $B(8,14)$. Find the coordinates of the point P which divides the interval AB internally in the ratio 1 : 3. 2

Question 11 continues on page 8

Marks

- f) Find the greatest coefficient in the expansion of $(2x + 3)^9$. 3
- g) How many ways can 12 people be seated in a circle if 2 particular people must sit apart from each other? 2

Question 12 (15 marks) Start a new booklet

Marks

- a) Find the coefficient of x^2 in the expansion of $\left(x^2 - \frac{3}{x}\right)^7$ 2
- b) Solve the inequality $\frac{4}{x-2} \leq 1$. 2
- c) Use mathematical induction to prove that $7^n - 3^n$ is divisible by 4 for all positive integers $n \geq 1$. 3
- d) i) Show that $2 \sin x = x$ has a root between $x = 1$ and $x = 2$. 1
 ii) Taking $x = 1.8$ as an approximation for the solution of $2 \sin x = x$, use Newton's Method once to give a better approximation (1 decimal place). 2
- e) An archer hits a target on average 3 out of every 5 times she shoots. Find the probability that in 10 shots at the target: 1
 i) she hits it exactly once (3 significant figures)
 ii) She hits it at least 2 times (3 significant figures) 2
- f) Find the value of the constants a and b if $x^2 + x - 6$ is a factor of the polynomial $x^3 + 5x^2 + ax + b$. 2

Question 13 (15 marks) Start a new booklet

Marks

- a) When a particle is x metres from the origin, its velocity, v m/s, is given by 2

$$v = \sqrt{8x - 2x^2}$$

 Find the acceleration when the particle is 2 m to the right of the origin.
- b) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. It is given that the chord PQ has equation $y = \left(\frac{p+q}{2}\right)x - apq$. 1
 i) Show that the gradient of the tangent at P is p . 2
 ii) Prove that if PQ passes through the focus, then the tangent at P is parallel to the normal at Q . 1
- c) i) State the domain and range of $y = 4 \sin^{-1}(1 - x)$ 2
 ii) Hence sketch $y = 4 \sin^{-1}(1 - x)$, clearly showing all essential features. 1

Question 13 continues on page 11

d) For the graph of $f(x) = \frac{x+1}{x^2+4}$

i) Find the coordinates of any stationary points and determine their nature. (1 DECIMAL PLACE)

ii) Find the horizontal asymptotes of $f(x) = \frac{x+1}{x^2+4}$

iii) Sketch the graph showing all essential features.

iv) By using the fact that $\frac{x+1}{x^2+4} = \frac{x}{x^2+4} + \frac{1}{x^2+4}$, or otherwise, show that the area bounded by $f(x) = \frac{x+1}{x^2+4}$, the x-axis and the lines $x = 0$ and $x = 2$ is equal to $\frac{1}{2} \left(\ln 2 + \frac{\pi}{4} \right) \text{ units}^2$.

2

1

1

3

Question 14 (15 marks) Start a new booklet

2

a) Prove that $\sin^{-1} \frac{1}{\sqrt{5}} + \sin^{-1} \frac{1}{\sqrt{10}} = \frac{\pi}{4}$

b) A particle is projected from a point O with an initial velocity $v \text{ m/s}$ and with an angle of projection α , where $0 \leq \alpha \leq 90^\circ$ and where $g \text{ m/s}^2$ is the acceleration due to gravity. Under these conditions you may assume that the equations for the horizontal and vertical displacements at time t are given by:

$$x = v t \cos \alpha \quad y = v t \sin \alpha - \frac{1}{2} g t^2$$

i) Prove that $y = x \tan \alpha - \frac{g x^2}{2 v^2} \sec^2 \alpha$.

2

ii) Find the time of flight and the range in terms of v , α and g .

2

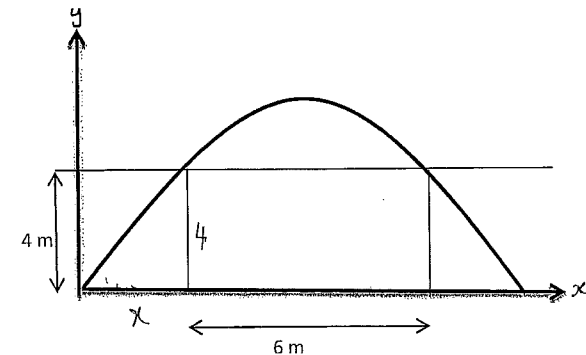
iii) If R is the range of the projectile on the horizontal plane, prove that:

2

$$y = x \left(1 - \frac{x}{R} \right) \tan \alpha$$

iv) If $\alpha = 45^\circ$ and the particle just clears two walls 6 m apart, both at a height of 4 m, find the range of the projectile, R .

3



Question 14 continues on page 13

Marks

c) i) Write down the expansion of $(1+x)^{2n}$

1

ii) Prove that $2^{2n} = \sum_{k=0}^{2n} \binom{2n}{k}$

3

iii) Prove that $\sum_{k=0}^n \binom{2n}{k} = 2^{2n-1} + \frac{(2n)!}{2(n!)^2}$

End of examination

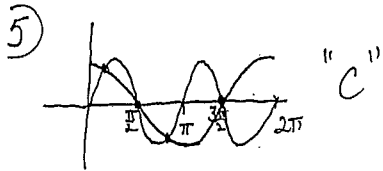
SECTION 1

① "b" ② $y = (p+q)^2 + 2pq$

$$y = x^2 - 2$$

"d"

③ "b" ④ "d"



⑥ $V = x^3$ $\frac{dV}{dx} = 3x^2$ $\frac{dV}{dt} = \frac{dV}{dx} \cdot \frac{dx}{dt}$
 $= 3(4)^2 \cdot (-0.5)$
 $= -24 \text{ cm}^3/\text{min}$
 "a"

⑦ No. of ways = $\frac{8!}{3!}$ "C"

⑧ $\alpha + \beta + \gamma = -\frac{8}{2}$ $\alpha + \beta = -\frac{5}{2}$
 $= -4$

$$\alpha^2 + \beta^2 + \gamma^2 = (\sum \alpha)^2 - 2(\sum \alpha\beta)$$

$$= (-4)^2 - 2\left(-\frac{5}{2}\right)$$

$$= 21$$

"a"

⑨ "b"

⑩ $x+1 + x+2 > 7-2x$ $x+1+7-2x > x+2$
 $4x > 4$ $2x < 6$
 $x > 1$ $x < 3$

$x+2 + 7-2x > x+1$ $x+1 > 0$ $x+2 > 0$
 $2x < 8$ $x > -1$ $x > -2$
 $x < 4$ $7-2x > 0$
 $\therefore 1 < x < 3$ "a" $2x < 7$ $x < 3\frac{1}{2}$

SECTION 2

⑪ a) $y = 3 \cos^{-1} \frac{x}{2}$

$$y' = \frac{-3}{\sqrt{1-x^2/4}} \cdot \frac{1}{2}$$

$$= \frac{-3}{\sqrt{4-x^2}}$$

$$\frac{x=1}{m = \frac{-3}{\sqrt{3}}}$$

$$= -\sqrt{3}$$

b) $I = \int \frac{x}{\sqrt{1-x^4}} dx$
 $u = x^2$
 $du = 2x dx$

$$I = \frac{1}{2} \int \frac{1}{\sqrt{1-u^2}} du$$

$$= \frac{1}{2} \sin^{-1}(u) + C$$

$$= \frac{1}{2} \sin^{-1}(x^2) + C$$

c) No. of selections = ${}^9C_3 {}^6C_2$
 $= 1260$

d) $I = \int_0^{\frac{3}{2}} \frac{2}{\sqrt{9-4x^2}} dx$
 $= \frac{2}{2} \int_0^{\frac{3}{2}} \frac{1}{\sqrt{\frac{9}{4}-x^2}} dx$
 $= \left[\sin^{-1} \frac{2x}{3} \right]_0^{\frac{3}{2}}$
 $= \sin^{-1} 1 - \sin^{-1} 0$
 $= \pi/2$

e) $P(x,y) = \left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n} \right)$
 $= \left(\frac{1.8+3(-4)}{1+3}, \frac{1(14)+3(6)}{1+3} \right)$
 $= (-1, 8)$

f) $\frac{T_{k+1}}{T_k} = \frac{n-k+1}{k} \cdot \frac{b}{a}$
 $= \frac{9-k+1}{k} \cdot \frac{3}{2}$
 $= \frac{30-3k}{2k}$

GREATEST COEFF $\frac{T_{k+1}}{T_k} \geq 1$

$$\frac{30-3k}{2k} \geq 1$$

$$5k \leq 30$$

$$k \leq 6$$

$\therefore T_6$ AND T_7 HAVE GREATEST COEFF.

$$T_7 = {}^9C_6 (2x)^3 (3)^6$$

$$= 489888x^3$$

$\therefore$ GREATEST COEFF = 489888

g) No. of ways = $10!2!$
 TOGETHER

No. of ways APART = $11! - 10!2!$
 $= 32659200$

a) $T_{k+1} = {}^nC_k a^{n-k} b^k$
 $= {}^7C_k (x^2)^{7-k} (-3x^{-1})^k$

TERM IN $x^2 = x^{14-2k} \cdot x^{-k}$
 $= x^{14-3k}$

$$\therefore 14-3k = 2$$

$$3k = 12$$

$$k = 4$$

$\therefore$ COEFF = ${}^7C_4 (-3)^4$
 $= 2835$

b) $\frac{4}{x-2} \leq 1$

$$4(x-2) \leq (x-2)^2$$

$$(x-2)^2 - 4(x-2) \geq 0$$

$$(x-2)[x-2-4] \geq 0$$

$$(x-2)(x-6) \geq 0$$

$$x < 2, x \geq 6$$

c) PROVE TRUE FOR $n=1$

$$7^1 - 3^1 = 4$$

$$= 4 \times 1$$

$\therefore$ DIVISIBLE BY 4 AND TRUE FOR $n=1$

ASSUME TRUE FOR $n=k$

$$7^k - 3^k = 4Q \text{ (Q SOME INTEGER)}$$

PROVE TRUE FOR $n=k+1$

$$7^{k+1} - 3^{k+1} = 4m \text{ (m SOME INTEGER)}$$

$$\text{LHS} = 7(7^k) - 3(3^k)$$

$$= 7(4Q + 3^k) - 3(3^k)$$

$$\begin{aligned}
 &= 28Q + 4(3^k) \\
 &= 4(7Q + 3^k) \\
 &= 4M \text{ (WHERE } M=7Q+3^k \text{ SOME INTEGER)} \\
 &= RHS
 \end{aligned}$$

$\therefore$ IF TRUE FOR $n=k$, THEN PROVED TRUE FOR $n=k+1$. BUT TRUE FOR $n=1$, $\therefore$ TRUE FOR $n=2$, AND BY PRINCIPLES OF INDUCTION, TRUE FOR ALL $n \geq 1$.

d) i) $2\sin x = x$ $f(x) = 2\sin x - x$
 $2\sin x - x = 0$
 $f(1) = 2\sin 1 - 1$ $f(2) = 2\sin 2 - 2$
 $= 0.6829...$ $= -0.1814...$

SINCE $f(1)$ AND $f(2)$ CHANGE SIGN, THERE IS A ROOT OF $f(x)=0$ BETWEEN $x=1$ AND $x=2$.

ii) $f'(x) = 2\cos x - 1$
 $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$
 $x_2 = 1.8 - \frac{2\sin 1.8 - 1.8}{2\cos 1.8 - 1}$
 $= 1.90155...$
 $= 1.9$ (1 DECIMAL PLACE)

e) i) $P(X=r) = {}^nC_r q^{n-r} p^r$ $p = \frac{3}{5}$ $q = \frac{2}{5}$

i) $P(X=1) = {}^{10}C_1 \left(\frac{2}{5}\right)^9 \left(\frac{3}{5}\right)^1$
 $= 0.00157$ (3 SIG FIG)

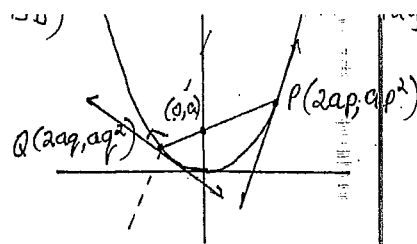
ii) $P(X \geq 2) = 1 - P(X=0) - P(X=1)$
 $= 1 - \left(\frac{2}{5}\right)^{10} - 0.00157$
 $= 0.998$

f) $(x+3)(x-2)$ IS A FACTOR
 $\therefore P(-3) = -27 + 45 - 3a + b = 0$ ①
 $P(2) = 8 + 20 + 2a + b = 0$ ②

$$\begin{aligned}
 3a - b &= 18 \quad \text{①} \\
 2a + b &= -28 \quad \text{②} \\
 \hline
 5a &= -10 \\
 a &= -2
 \end{aligned}$$

IN ①
 $-6 - b = 18$
 $b = -24$
 $\therefore a = -2, b = -24$

13) a) $V = \sqrt{8x - 2x^2}$
 $V^2 = 8x - 2x^2$
 $\frac{1}{2}V^2 = 4x - x^2$
 $\frac{d(\frac{1}{2}V^2)}{dx} = 4 - 2x$
 $\frac{x}{2} = 4 - 4$
 $= 0 \text{ ms}^{-2}$



i) $y = \frac{x^2}{4a}$
 $y' = \frac{2x}{4a}$
 $= \frac{x}{2a}$

at $P(2ap, ap^2)$
 $m = \frac{2ap}{2a}$
 $= p$

ii) $(0, a)$ SATISFIES $y = \left(\frac{p+q}{2}\right)x + apq$
 $\therefore a = \left(\frac{p+q}{2}\right) \cdot 0 + apq$
 $a = -apq$
 $pq = -1$
 $q = -\frac{1}{p}$

BUT q IS THE GRADIENT OF THE TANGENT AT Q

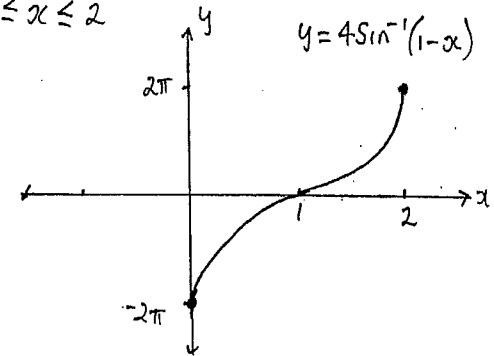
$\therefore m_{\text{NORMAL AT Q}} = \frac{-1}{-\frac{1}{p}}$
 $= p$

$\therefore$ NORMAL AT Q IS $\parallel$ TO TANGENT AT P .

$y = 4\sin(1-x)$

DOMAIN RANGE

$-1 \leq 1-x \leq 1$ $-2\pi \leq y \leq 2\pi$
 $-2 \leq -x \leq 0$
 $0 \leq x \leq 2$



d) $f(x) = \frac{x+1}{x^2+4}$
 $f'(x) = \frac{(x^2+4) - (x+1)(2x)}{(x^2+4)^2}$
 $= \frac{x^2+4-2x^2-2x}{(x^2+4)^2}$
 $= \frac{-x^2-2x+4}{(x^2+4)^2}$

STAT PTS $f'(x) = 0$
 $\frac{-x^2-2x+4}{(x^2+4)^2} = 0$

$x^2+2x-4 = 0$

$x = \frac{-2 \pm \sqrt{20}}{2}$

$= -1 + \sqrt{5}, -1 - \sqrt{5}$
 $= 1.2, -3.2$

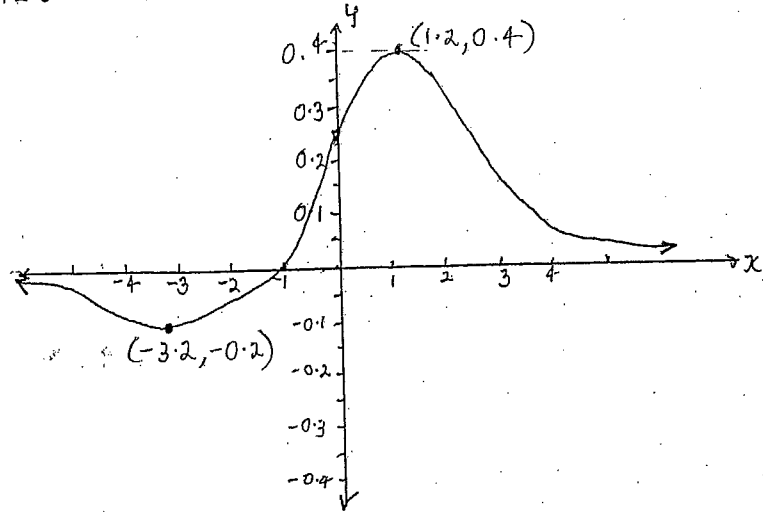
ii) VERTICAL ASYMPTOTES WHEN $x^2+4=0$
 $\therefore$ NO VERTICAL ASYMPTOTES

$$\lim_{x \rightarrow \infty} \frac{x+1}{x^2+4} = 0^+$$

$$\lim_{x \rightarrow -\infty} \frac{x+1}{x^2+4} = 0^-$$

$\therefore y=0$ IS A HORIZONTAL ASYMPTOTE

iii)



$$\begin{aligned} \text{iv) } A &= \int_0^2 \frac{x+1}{x^2+4} dx \\ &= \int_0^2 \frac{x}{x^2+4} dx + \int_0^2 \frac{1}{x^2+4} dx \\ &= \frac{1}{2} \left[\ln(x^2+4) \right]_0^2 + \frac{1}{2} \left[\tan^{-1} \frac{x}{2} \right]_0^2 \\ &= \frac{1}{2} [\ln 8 - \ln 4] + \frac{1}{2} [\tan^{-1} 1 - \tan^{-1} 0] \\ &= \frac{1}{2} \left[\ln 2 + \frac{\pi}{4} \right] \cdot v^2 \end{aligned}$$

x	1	1.2	2
f'(x)	+	0	-ve

MAXIMUM TURNING POINT AT (1.2, 0.4)

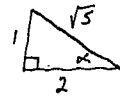
x	-4	-3.2	-3
f'(x)	-ve	0	+

MINIMUM TURNING POINT AT (-3.2, -0.2)

$$\text{a) } \sin^{-1} \frac{1}{\sqrt{5}} + \sin^{-1} \frac{1}{\sqrt{10}} = \frac{\pi}{4}$$

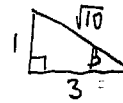
$$\sin^{-1} \frac{1}{\sqrt{5}} = \alpha$$

$$\sin \alpha = \frac{1}{\sqrt{5}}$$



$$\sin^{-1} \frac{1}{\sqrt{10}} = \beta$$

$$\sin \beta = \frac{1}{\sqrt{10}}$$



$$\begin{aligned} \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\ &= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}} \end{aligned}$$

$$\text{ie } \tan(\alpha + \beta) = 1$$

$$\therefore \alpha + \beta = \frac{\pi}{4}$$

$$\text{b) i) } x = vt \cos \alpha \quad y = vt \sin \alpha - \frac{1}{2} g t^2$$

$$t = \frac{x}{v \cos \alpha}$$

$$y = v \sin \alpha \left(\frac{x}{v \cos \alpha} \right) - \frac{1}{2} g \left(\frac{x^2}{v^2 \cos^2 \alpha} \right) \therefore y = x \left(1 - \frac{x}{R} \right) \tan \alpha$$

$$y = x \tan \alpha - \frac{g x^2 \sec^2 \alpha}{2 v^2}$$

iv)

$$\text{ii) } t_{\text{FLIGHT}}, y = 0$$

$$v t \sin \alpha - \frac{1}{2} g t^2 = 0$$

$$t(v \sin \alpha - \frac{g t}{2}) = 0$$

$$t = 0, \frac{2 v \sin \alpha}{g}$$

$$\therefore t_{\text{FLIGHT}} = \frac{2 v \sin \alpha}{g}$$

$$x_{\text{RANGE}} = v \cos \alpha \left(\frac{2 v \sin \alpha}{g} \right)$$

$$= \frac{v^2 \sin 2\alpha}{g}$$

iii)

$$R = \frac{v^2 \sin 2\alpha}{g}$$

$$\therefore v^2 = \frac{g R}{\sin 2\alpha}$$

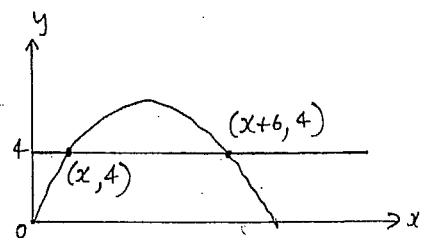
$$\text{SUBST IN } y = x \tan \alpha - \frac{g x^2 \sec^2 \alpha}{2 v^2}$$

$$y = x \tan \alpha - \frac{g x^2 \sin 2\alpha}{2 g R \cos^2 \alpha}$$

$$= x \tan \alpha - \frac{x^2 2 \sin \alpha \cos \alpha}{2 R \cos \alpha \cos \alpha}$$

$$= x \tan \alpha - \frac{x^2 \tan \alpha}{R}$$

14b)
iv)



SUB $(x, 4)$ IN $y = x(1 - \frac{x}{R}) \tan \alpha$

$$4 = x(1 - \frac{x}{R}) \tan 45^\circ$$

$$4 = x(1 - \frac{x}{R}) \quad (1)$$

SUB $(x+6, 4)$ IN $y = x(1 - \frac{x}{R}) \tan \alpha$

$$4 = (x+6)(1 - \frac{x+6}{R}) \tan 45^\circ$$

$$4 = (x+6)(1 - \frac{x+6}{R})$$

FROM (1)

MULTIPLY (1) BY R

$$4R = xR - x^2$$

$$x^2 = R(x-4)$$

$$R = \frac{x^2}{x-4}$$

INTO (2)

$$4 = (x+6)(1 - \frac{(x+6)(x-4)}{x^2})$$

$$\frac{4}{x+6} = \frac{x^2 - (x^2 + 2x - 24)}{x^2}$$

$$\frac{4}{x+6} = \frac{24 - 2x}{x^2}$$

$$4x^2 = 24x - 2x^2 + 144 - 12x$$

$$6x^2 - 12x - 144 = 0$$

$$x^2 - 2x - 24 = 0$$

$$(x-8)(x+6) = 0$$

$$x = 8, -6$$

$$\therefore x = 8 \quad (x > 0)$$

$$\therefore R = \frac{8^2}{8-4}$$

$$= 16 \text{ m}$$

c)

$$i) (1+x)^{2n} = \binom{2n}{0} + \binom{2n}{1}x + \binom{2n}{2}x^2 + \dots + \binom{2n}{n}x^n + \dots + \binom{2n}{2n}x^{2n}$$

ii) IN THE EXPANSION IN i)

LET $x = 1$

$$2^{2n} = \binom{2n}{0} + \binom{2n}{1} + \binom{2n}{2} + \dots + \binom{2n}{2n}$$

$$= \binom{2n}{0} + \binom{2n}{1} + \binom{2n}{2} + \dots + \binom{2n}{2n}$$

$$= \sum_{k=0}^{2n} \binom{2n}{k}$$

iii)

GIVEN

$$2^{2n} = \underbrace{\binom{2n}{0} + \binom{2n}{1} + \dots + \binom{2n}{n-1}}_{S_n} + \underbrace{\binom{2n}{n} + \binom{2n}{n+1} + \dots + \binom{2n}{2n}}_{S_{\text{LAST } n}}$$

S_{LAST n} TERMS

SINCE ${}^nC_k = {}^nC_{n-k}$, THEN DUE TO SYMMETRY

$$\therefore S_n = S_{\text{LAST } n}$$

$$\text{So } 2^{2n} = 2S_n + \binom{2n}{n}$$

$$2S_n = 2^{2n} - \frac{(2n)!}{n!n!}$$

$$S_n = \frac{2^{2n-1} - \frac{(2n)!}{2(n!)^2}}{1}$$

$$\therefore \sum_{k=0}^n \binom{2n}{k} = 2^{2n-1} - \frac{(2n)!}{2(n!)^2} + \frac{(2n)!}{(n!)^2}$$

$$= 2^{2n-1} + \frac{(2n)!}{2(n!)^2}$$